

RETAIL SALES PERFORMANCE DASHBOARD: BIG MART PRODUCT & OUTLET ANALYSIS USING MICROSOFT POWER BI

Internship Project Report

Submitted in Partial Fulfillment of the Requirements for the Internship Program

Submitted By: Tanya Sinha

Role: Power BI Intern

Domain: Power BI

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Internship Company

Sysslan IT Solutions

DECLARATION

I hereby declare that this internship project report entitled "**Retail Sales Performance Dashboard: Big Mart Product & Outlet Analysis using Microsoft Power BI**" is an original work carried out by me during my internship at **Sysslan IT Solutions**.

The work presented in this report has not been submitted previously for any degree, diploma, or certification at any institution. All sources of information used in this project have been properly acknowledged and cited.

I further declare that the analysis, dashboard development, and business insights presented in this report were developed by me using Microsoft Power BI under the guidance of my internship mentor.

Submitted By

Tanya Sinha

Abstract

The retail industry generates a significant amount of transactional data every day. Converting this raw data into meaningful business insights is essential for making informed decisions related to inventory management, product performance, outlet efficiency, and customer satisfaction. Business Intelligence (BI) tools help organizations transform complex datasets into interactive reports and dashboards that support strategic decision-making.

The objective of this internship project was to develop a **Retail Sales Performance Dashboard** using **Microsoft Power BI** based on the **Big Mart Sales Dataset**. The project involved multiple stages including data cleaning, data transformation, data modeling, DAX calculations, dashboard design, and business insight generation. Power Query was used to preprocess the data by handling missing values, standardizing inconsistent entries, and validating data quality before analysis.

Several interactive dashboards were developed during the project, including an Executive Dashboard, Product Analytics Dashboard, Outlet Analytics Dashboard, Business Insights Dashboard, and a Drill-Through Product Details page. These dashboards provide a comprehensive view of product categories, outlet performance, pricing trends, visibility analysis, and key performance indicators (KPIs).

The developed solution enables users to explore retail data through interactive filters, navigation buttons, drill-through functionality, dynamic titles, synchronized slicers, and data visualizations. Business recommendations were derived from the analysis to help improve product management, pricing strategies, and outlet performance. Overall, this project demonstrates the practical application of Business Intelligence concepts using Microsoft Power BI and highlights the importance of data-driven decision-making in the retail industry.

Keywords: Microsoft Power BI, Business Intelligence, Retail Analytics, Dashboard, Data Visualization, DAX, Power Query, Sales Analysis

CHAPTER 1: COMPANY PROFILE

1.1 About the Company

Sysslan IT Solutions is a technology-driven organization that provides innovative software solutions in the domains of **Cloud Computing, Artificial Intelligence (AI), Machine Learning (ML), Blockchain Technology, Data Analytics, and Business Intelligence**. The company focuses on developing modern technology solutions that help businesses improve operational efficiency and make better decisions through data.

The organization provides internship opportunities to students to enhance their practical knowledge by working on real-world projects. During the internship, students gain exposure to industry-standard tools, project development methodologies, and modern business technologies.

The internship program emphasizes practical learning through project-based assignments, enabling interns to strengthen their analytical, technical, and problem-solving skills.

1.2 Vision

To empower organizations with innovative technology solutions that promote digital transformation and data-driven decision-making.

1.3 Mission

- Deliver high-quality software and analytical solutions.
- Encourage innovation through emerging technologies.
- Provide practical industry exposure to students.
- Promote Business Intelligence and Data Analytics solutions.

1.4 Services Offered

- Artificial Intelligence (AI)
- Machine Learning (ML)
- Cloud Computing
- Blockchain Technology
- Business Intelligence
- Data Analytics
- Dashboard Development
- Software Development

CHAPTER 2: PROJECT INTRODUCTION

2.1 Project Overview

The **Retail Sales Performance Dashboard** is a Business Intelligence solution developed using Microsoft Power BI to analyze retail sales data from the Big Mart dataset. The dashboard provides a centralized platform for monitoring product performance, outlet performance, pricing analysis, and key business metrics through interactive visualizations.

The project was developed as part of the internship program at **Sysslan IT Solutions** with the objective of applying Business Intelligence concepts to solve a real-world retail analytics problem.

The dashboard consists of five interactive pages:

1. Executive Dashboard
2. Product Analytics Dashboard
3. Outlet Analytics Dashboard
4. Business Insights Dashboard
5. Product Details (Drill Through)

Each dashboard focuses on a specific business perspective, enabling decision-makers to explore retail performance through interactive filters, navigation buttons, dynamic titles, synchronized slicers, and drill-through functionality.

2.2 Need of the Project

Retail organizations manage thousands of products across multiple outlets. Without proper visualization tools, it becomes difficult to monitor product performance, outlet efficiency, and pricing trends.

The developed dashboard addresses these challenges by providing:

- Interactive KPI monitoring
- Product category analysis
- Outlet performance evaluation
- Pricing analysis
- Business recommendations
- Decision support through visual analytics

This solution helps managers understand business performance quickly and supports data-driven decision-making.

CHAPTER 3: PROBLEM STATEMENT

3.1 Problem Statement

In today's retail industry, businesses generate a massive number of sales and inventory data every day. This data contains valuable information about product performance, customer preferences, outlet efficiency, pricing strategies, and inventory management. However, without proper analysis and visualization, it becomes difficult for managers to extract meaningful insights from raw data.

Traditional reporting methods, such as spreadsheets and static reports, are time-consuming and often fail to provide real-time business insights. Decision-makers require an interactive and dynamic solution that allows them to monitor key performance indicators (KPIs), compare product categories, analyze outlet performance, and identify business opportunities quickly.

The Big Mart retail dataset contains information about different products, outlet characteristics, product visibility, pricing, and sales-related attributes. However, the dataset in its raw form includes missing values, inconsistent data entries, and requires transformation before meaningful analysis can be performed.

Therefore, there is a need to develop an interactive Business Intelligence dashboard that can transform raw retail data into visual insights, helping business users make informed decisions regarding product management, outlet performance, and overall business growth.

The developed Power BI dashboard addresses these challenges by integrating data cleaning, data transformation, DAX calculations, and interactive visualizations into a single reporting solution.

3.2 Existing System

The existing system for retail data analysis primarily relies on spreadsheets, manual reports, and traditional tabular data. While these methods store business information, they have several limitations when handling large datasets.

Limitations of the Existing System

- Manual analysis is time-consuming.
- Reports are static and cannot be filtered dynamically.
- Difficult to identify trends and patterns.
- No interactive visualization.
- Limited support for business decision-making.
- High chances of human error during reporting.
- Difficult to compare multiple business parameters simultaneously.

These limitations reduce the efficiency of business analysis and slow down the decision-making process.

3.3 Proposed System

The proposed system is an interactive Retail Sales Performance Dashboard developed using Microsoft Power BI.

The dashboard integrates multiple Business Intelligence features such as:

- Data Cleaning using Power Query
- Data Transformation
- DAX Measures
- Interactive KPI Cards
- Product Analytics
- Outlet Analytics
- Business Insights
- Drill-Through Analysis
- Dynamic Titles
- Navigation Buttons
- Interactive Filters and Slicers

The dashboard enables managers to monitor retail performance through real-time visualizations and interactive reports.

3.4 Benefits of the Proposed System

The proposed dashboard offers the following benefits:

- Interactive business reporting
- Faster decision-making
- Better understanding of product performance
- Easy comparison of outlet performance
- Dynamic filtering of business data
- Improved visualization of KPIs
- Reduced manual reporting effort
- Better inventory and pricing analysis
- Enhanced business intelligence capabilities

CHAPTER 4: PROJECT OBJECTIVES

4.1 General Objective

The primary objective of this project is to develop an interactive Retail Sales Performance Dashboard using Microsoft Power BI that transforms raw retail data into meaningful business insights through data visualization and Business Intelligence techniques.

4.2 Specific Objectives

The specific objectives of the project are:

- To clean and preprocess the retail sales dataset using Power Query.
- To handle missing values and standardize inconsistent data.
- To create reusable DAX measures for business calculations.
- To design interactive KPI cards for business monitoring.
- To analyze product performance using various visualizations.
- To analyze outlet performance based on outlet type, location, and size.
- To develop an Executive Dashboard for business overview.
- To build Product Analytics and Outlet Analytics dashboards.
- To generate business insights and recommendations.
- To implement advanced Power BI features such as Drill Through, Dynamic Titles, Navigation Buttons, and Sync Slicers.
- To support data-driven business decision-making.

4.3 Scope of the Project

The scope of this project includes the complete analysis of retail sales data using Microsoft Power BI.

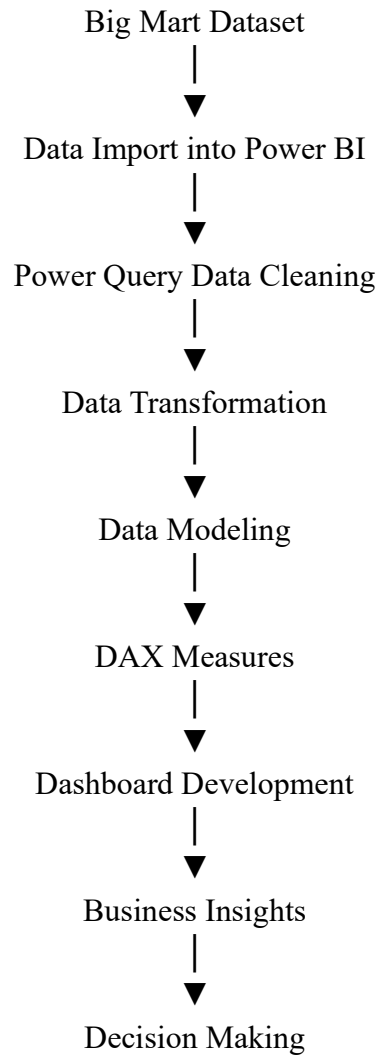
The project covers:

- Data Import
- Data Cleaning
- Data Transformation
- Data Modeling
- DAX Calculations
- Dashboard Development
- Product Analysis
- Outlet Analysis

- Business Insights
- Interactive Reporting

The developed dashboard is intended for business managers, analysts, and decision-makers who require quick access to retail performance metrics.

4.4 Project Workflow



CHAPTER 5: DATASET DESCRIPTION

5.1 Dataset Overview

The dataset used in this project is the Big Mart Sales Dataset, which contains information related to products and outlets. It is widely used for retail sales analysis and Business Intelligence projects.

The dataset was imported into Microsoft Power BI and processed using Power Query before visualization.

Dataset Summary

Particular	Details
Dataset Name	Big Mart Sales Dataset
Total Records	5681
Total Columns	11
Tool Used	Microsoft Power BI
Data Source	CSV File

5.2 Dataset Attributes

Column Name	Description
Item_Identifier	Unique Product ID
Item_Weight	Weight of Product
Item_Fat_Content	Fat Category (Low Fat/Regular)
Item_Visibility	Shelf Visibility
Item_Type	Product Category
Item_MRP	Maximum Retail Price
Outlet_Identifier	Outlet ID
Outlet_Establishment_Year	Year of Establishment
Outlet_Size	Size of Outlet
Outlet_Location_Type	Tier of Outlet
Outlet_Type	Type of Outlet

5.3 Dataset Statistics

- Total Records: 5681
- Total Columns: 11
- Missing Values: Item_Weight, Outlet_Size
- Duplicate Records: Not Found
- Unique Fat Categories: 2 (Low Fat, Regular)

CHAPTER 6: DATA CLEANING & TRANSFORMATION

6.1 Data Cleaning Process

Before analysis, the dataset was cleaned using Power Query.

The following operations were performed:

- Missing values were identified.
- Missing values in Item_Weight were replaced appropriately.
- Missing values in Outlet_Size were handled.
- Inconsistent values in Item_Fat_Content were standardized into Low Fat and Regular.
- Duplicate records were checked.
- Data types were verified and corrected where required.

6.2 Power Query Operations

The following Power Query features were used:

- Remove Errors
- Replace Values
- Change Data Type
- Filter Rows
- Rename Columns
- Close & Apply

6.3 Data Cleaning Outcome

After preprocessing, the dataset became consistent and ready for dashboard development. This improved the accuracy of DAX calculations and ensured reliable business insights.

CHAPTER 7: DATA MODELING & DAX MEASURES

7.1 Data Modeling

After data cleaning, the dataset was loaded into Microsoft Power BI for analysis. Since this project used a single cleaned dataset, no additional relationships were required. The data model served as the foundation for creating calculations, KPIs, and interactive dashboards.

7.2 DAX (Data Analysis Expressions)

DAX (Data Analysis Expressions) is a formula language used in Microsoft Power BI to perform calculations and create custom measures. It helps generate dynamic values based on filters and user selections.

In this project, DAX was used to calculate KPIs, dynamic titles, and business metrics.

7.3 DAX Measures Used

Measure	Purpose
Total Records	Displays the total number of records
Total Outlets	Counts the total number of outlets
Total Item Types	Counts unique product categories
Average MRP	Calculates average product price
Highest MRP	Finds the highest product price
Lowest MRP	Finds the lowest product price
Average Weight	Calculates average product weight
Average Visibility	Calculates average product visibility
Selected Item Type	Displays selected product category
Selected Outlet Type	Displays selected outlet type
Dynamic Title Measures	Updates chart titles based on filters

CHAPTER 8: DASHBOARD DEVELOPMENT

8.1 Dashboard Overview

The Power BI report consists of five interactive dashboard pages, each designed to answer specific business questions. The dashboards provide a complete analysis of product performance, outlet performance, and overall retail business trends.

The report includes:

- Executive Dashboard
- Product Analytics Dashboard
- Outlet Analytics Dashboard
- Business Insights Dashboard
- Product Details (Drill Through)

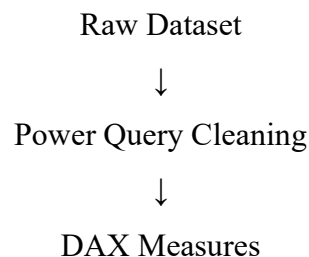
8.2 Dashboard Features

The following features were implemented during dashboard development:

- KPI Cards
- Interactive Charts
- Slicers and Filters
- Navigation Buttons
- Dynamic Titles
- Drill Through
- Sync Slicers
- Edit Interactions
- Conditional Formatting

These features improve user experience and make the report interactive and easy to analyze.

8.3 Dashboard Workflow





Interactive Dashboards



Business Insights



Decision Making

CHAPTER 9: DASHBOARD ANALYSIS

9.1 Executive Dashboard

The Executive Dashboard provides a summary of the complete retail dataset through interactive KPIs and visualizations. It helps business managers monitor overall retail performance without exploring detailed reports. The dashboard includes important KPIs such as Total Products, Total Outlets, Average MRP, Product Categories, Average Weight, and Average Visibility. Interactive slicers allow users to filter the report by Item Type, Outlet Type, Outlet Size, Location Type, and Fat Content.

Major Visualizations

- Products by Category
- Product Mix by Fat Content
- Average MRP by Product Category
- Products by Outlet Type
- Outlet Size Distribution
- Outlet Establishment Trend

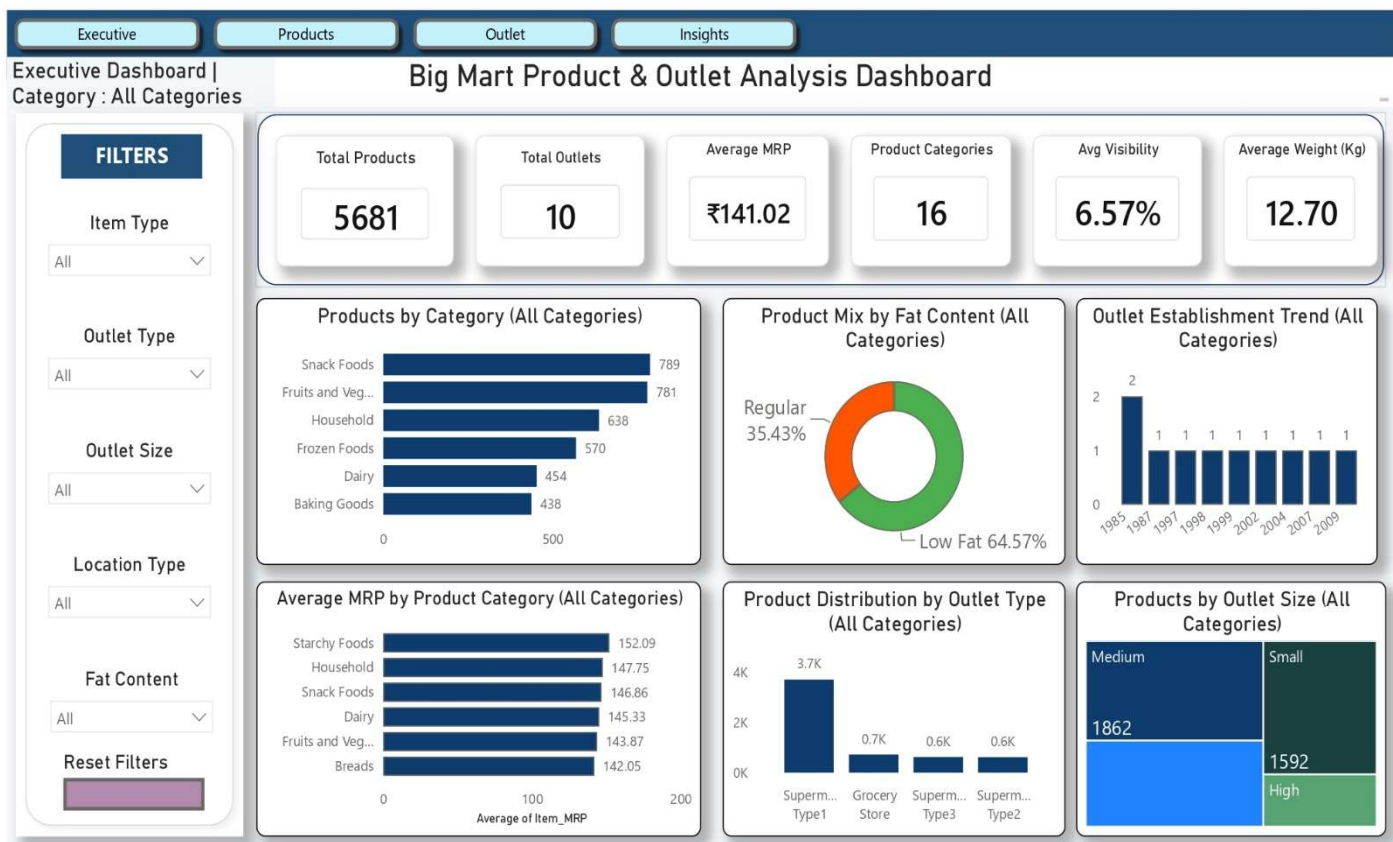


Figure 9.1

9.2 Product Analytics Dashboard

The Product Analytics Dashboard focuses on analyzing product-related information. It helps identify product categories with high prices, better visibility, and higher product availability.

The dashboard displays KPIs such as Highest MRP, Lowest MRP, Average MRP, Average Weight, and Average Visibility. Various charts are used to compare product categories and understand their distribution.

Major Visualizations

- Average MRP by Product Category
- Average Weight by Product Category
- Average Shelf Visibility
- Product Distribution by Fat Content
- Top Product Categories
- Product Details Matrix

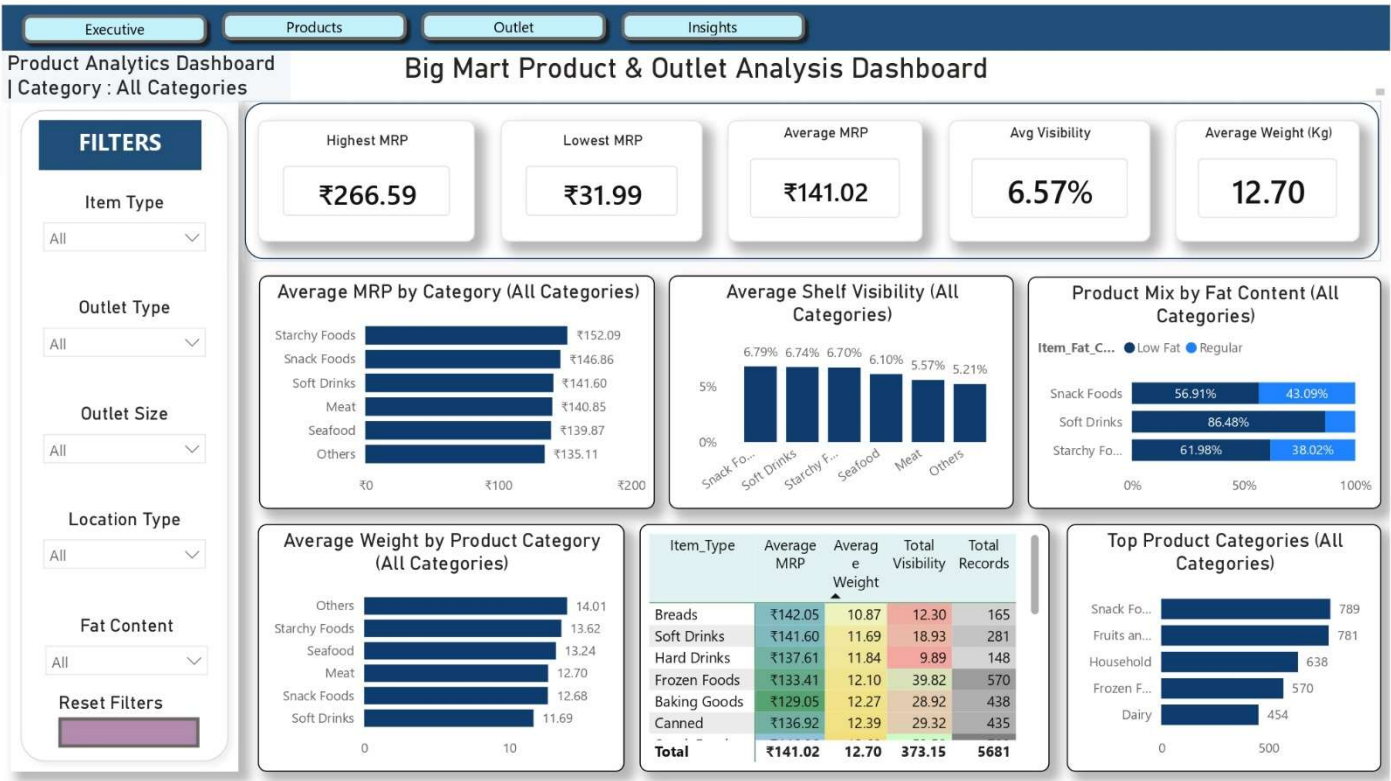


Figure 9.2

9.3 Outlet Analytics Dashboard

The Outlet Analytics Dashboard provides insights into outlet performance. It allows users to compare different outlet types, outlet sizes, and outlet locations.

The dashboard includes KPIs such as Total Outlets, Oldest Outlet, Newest Outlet, Average Outlet Age, and Total Locations.

Major Visualizations

- Products by Outlet Type
- Average MRP by Outlet Type
- Outlet Distribution by Location Type
- Outlet Size Distribution
- Outlet Establishment Trend
- Outlet Details Matrix

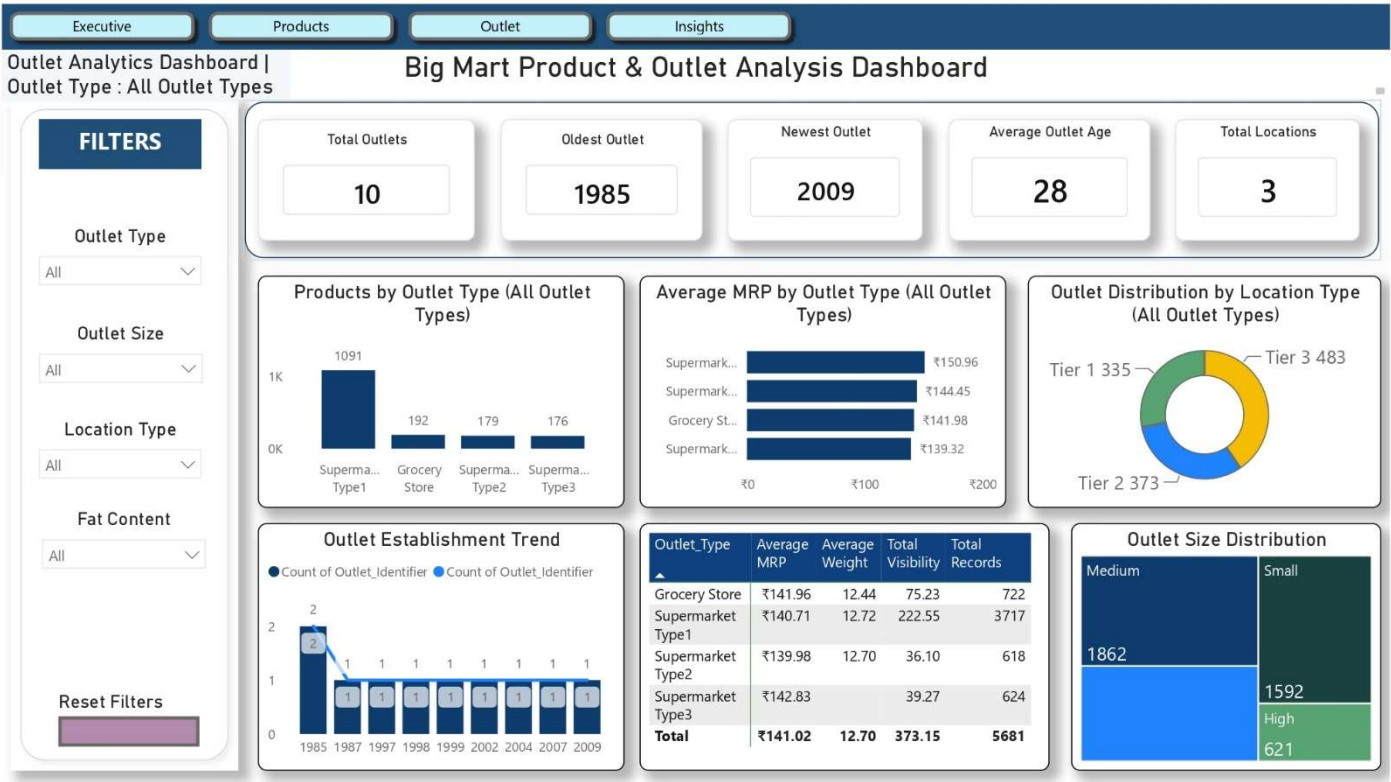


Figure 9.3

9.4 Business Insights Dashboard

The Business Insights Dashboard summarizes the overall findings of the analysis. It converts analytical results into business-friendly insights and recommendations.

The dashboard includes:

- Dataset Summary
- Key Business Insights
- Business Recommendations
- Skills & Technologies Used
- Project Outcome

This page helps management quickly understand the overall performance and identify areas for improvement.

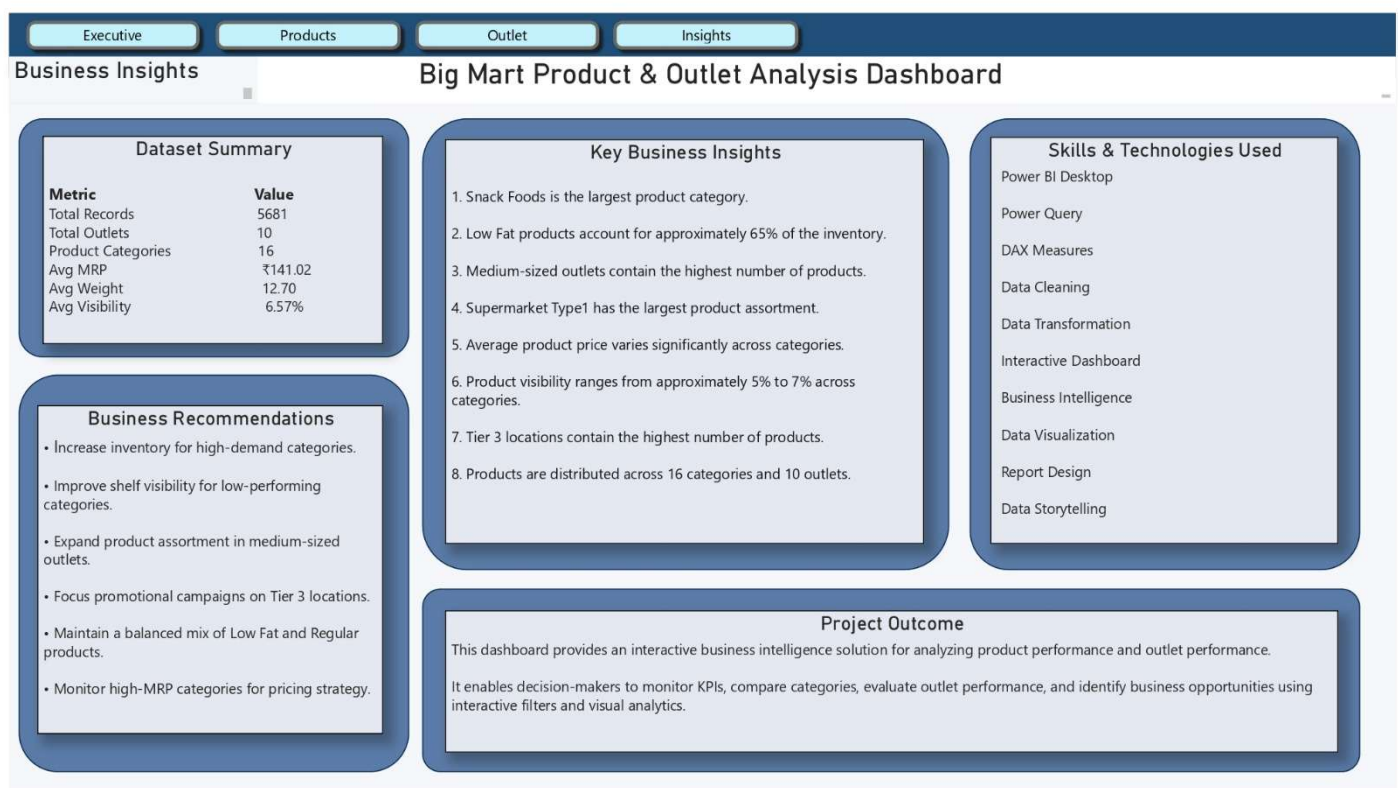


Figure 9.4

9.5 Product Details (Drill Through)

A Drill Through page was created to provide detailed information about a selected product category. Users can right-click on a product category from the dashboard and navigate to this page for deeper analysis.

The page includes:

- Highest MRP
- Lowest MRP
- Average MRP
- Average Weight
- Average Visibility
- Product Details Table
- Category-wise Charts

A Back button is provided to return to the previous dashboard.

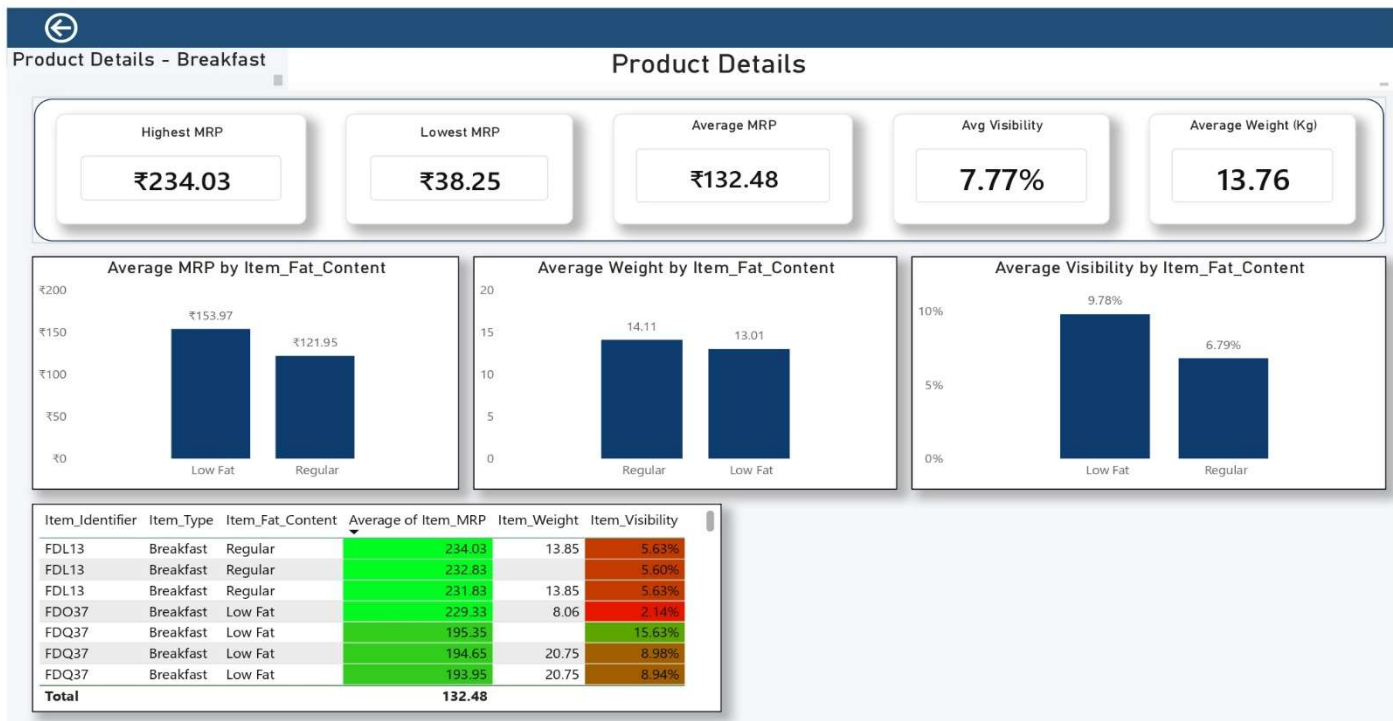


Figure 9.5:

CHAPTER 10: ADVANCED POWER BI FEATURES

To improve usability and make the dashboard interactive, several advanced Power BI features were implemented.

10.1 Navigation Buttons

Navigation buttons were added to allow users to switch between dashboard pages without using the default page tabs.

10.2 Dynamic Titles

Dynamic titles were created using DAX measures. The chart titles automatically update according to the selected filters, making the report easier to understand.

10.3 Sync Slicers

Sync Slicers were used to maintain the same filter selection across multiple dashboard pages, ensuring a consistent user experience.

10.4 Edit Interactions

Edit Interactions were configured to control how one visual affects another. This improves report readability and prevents unnecessary filtering.

10.5 Drill Through

Drill Through functionality allows users to move from summary-level dashboards to detailed product-level analysis.

10.6 Conditional Formatting

Conditional formatting was applied to the matrix visual to improve readability and highlight important values.

10.7 Interactive Filters

Dropdown slicers were used to filter data based on:

- Item Type
- Fat Content
- Outlet Type
- Outlet Size
- Outlet Location Type

10.8 Benefits of Advanced Features

The implemented features provide the following advantages:

- Improved navigation
- Better user experience
- Interactive data exploration
- Faster business analysis
- Enhanced decision-making

CHAPTER 11: BUSINESS INSIGHTS

The dashboard analysis provided several valuable insights that can help management understand the current retail performance and support future business decisions.

11.1 Key Findings

- Snack Foods emerged as one of the largest product categories in terms of product count.
- Low Fat products contributed a major share of the available inventory.
- Supermarket Type1 contained the highest number of products among all outlet types.
- Medium-sized outlets showed better product availability compared to small outlets.
- Average MRP varied significantly across different product categories.
- Product visibility differed among categories, indicating opportunities to improve shelf placement.
- Tier-based outlet analysis showed differences in product distribution across locations.
- Interactive dashboards enabled quick comparison of products and outlets using filters and slicers.

11.2 Business Value

The developed dashboard helps management to:

- Monitor important KPIs in one place.
- Compare product performance across categories.
- Evaluate outlet performance.
- Identify high-value and low-performing products.
- Support faster and more accurate business decisions.

CHAPTER 12: RECOMMENDATIONS

Based on the analysis, the following recommendations are suggested:

1. Increase inventory for high-performing product categories to meet customer demand.
2. Improve shelf visibility for products with low visibility to increase customer attention.
3. Review pricing strategies for high MRP products to maintain market competitiveness.
4. Strengthen inventory planning for medium and high-performing outlets.
5. Use dashboard insights regularly for monitoring business performance and decision-making.
6. Continue updating the dashboard with new sales data to ensure accurate and real-time analysis.

CHAPTER 13: CHALLENGES FACED

During the development of this project, several challenges were encountered:

- Handling missing values in the dataset.
- Standardizing inconsistent values in the Item Fat Content column.
- Creating accurate DAX measures for KPI calculations.
- Designing a clean and user-friendly dashboard layout.
- Managing interactions between multiple visuals.
- Implementing advanced features such as Drill Through and Dynamic Titles.
- Maintaining consistent formatting across all dashboard pages.

These challenges were resolved using Power Query, DAX functions, and Power BI visualization tools.

CHAPTER 14: LEARNING OUTCOMES

This internship project helped in developing both technical and analytical skills.

Technical Skills

- Microsoft Power BI
- Power Query
- DAX (Data Analysis Expressions)
- Data Cleaning and Transformation
- Dashboard Design
- Interactive Visualizations

Analytical Skills

- Business Intelligence
- Retail Data Analysis
- KPI Development
- Data Storytelling
- Decision Support Analysis

Professional Skills

- Problem Solving
- Report Preparation
- Data Interpretation
- Presentation Skills
- Business Communication

CHAPTER 15: CONCLUSION

The Retail Sales Performance Dashboard was successfully developed using Microsoft Power BI as part of the internship at Sysslan IT Solutions.

The project demonstrated the complete Business Intelligence workflow, including data cleaning, transformation, DAX calculations, dashboard development, and business insight generation. Interactive dashboards enabled users to analyze product and outlet performance through KPIs, charts, slicers, and advanced Power BI features.

The developed solution simplifies retail data analysis and provides meaningful insights that can support better business decisions. Overall, the internship enhanced practical knowledge of Business Intelligence concepts and strengthened skills in Microsoft Power BI.